A missing factor of i: source-level verification of the hyper-invariant tensor network and of tensor network renormalization

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Hyper-invariant tensor networks [1] are a proposed holographic ansatz built from a uniform tiling of hyperbolic space, advertised as compatible with a variational use because their building blocks obey *multi-tensor* isometry constraints. We set out to *use* these blocks, and were forced to verify them first.

We report a source-level verification of the printed building blocks of arXiv:1704.04229v1 (the `Hyper.tex` source snapshot). **The 4×4 tensor Y that the source prints as its doubly-unitary example is not doubly unitary.** In the (12|34) grouping its Gram matrix is $YY^\dagger = \not\vdash$ except for a single non-vanishing entry, $(YY^\dagger)_{14} = (YY^\dagger)_{41} = 2c_1 s_1$, where $c_1 = \cos\theta_1$, $s_1 = \sin\theta_1$; at the reference point $\theta_1 = 0.37$ this is 6.742879×10^{-1}. Every other matrix in the same appendix (the R, Q and reflection tensors) is exactly unitary. The defect is removed by a *single* modification — replacing the four corner entries $s_1 \to i\, s_1$ — which we identify with an independently published tensor of Steinberg and Prior [2]; their tensor satisfies the source's multi-tensor constraints, while the printed one does not (residuals 2.697152 and 9.497650 for w and u, against 1.776357×10^{-15} and 3.552714×10^{-15} for the corrected tensor).

We then sharpen the statement into a structural one. **The contractions that define the two multi-tensor constraints can only "see" one grouping of Y**, the (12|34) partition: the co-isometry condition for w reduces to $WW^\dagger = \mathrm{Tr}(Y^\dagger Y)\,(P_{12}(Y_1) \otimes P_{12}(Y_2))$ with P_{12} the (12|34) Gram, and the u condition reduces to the *same* condition on Y, through a factor that is independent of the surrounding tensors. Consequently the alternative groupings (13|24) and (14|23) never enter, and the source's doubly-unitary requirement is *strictly stronger than necessary*. This yields a reusable admissibility criterion: Y is unitary on (12|34) $\iff$ the multi-tensor constraints hold $\iff$ the network is genuinely hyper-invariant.

As a second, independent case in the same genre we verify the appendix of arXiv:1412.0732 (`TNR.tex`): two of its factorisation identities are *mutually exclusive*, the first being impossible for any real tensor because it equates a positive-semidefinite matrix with one of minimum eigenvalue -2.

Finally, for completeness and as a boundary, we report the checks we ran on our own building blocks — MERA, which is exactly isometric by construction and whose negative control exposes a four-order-of-magnitude gap between the isometry residual and the downstream reading, and Gaussian cMERA, which passes its self-check 9/9 yet carries no discriminating power at the state level. **Erratum scope.** The finding reported here is strictly a *source-level literature erratum*: we identify a typographical defect in a specific tensor printed in arXiv:1704.04229v1. It is *not* a claim that the hyper-invariant tensor network *method* is flawed, nor that its *theoretical foundation* is incorrect. We test a concrete implementation artifact (one printed matrix), not the underlying theory.

Boundary. This is a source-level report about what is printed. We make no claim about any version other than the v1 arXiv source we read, and no claim about the intent or the correctness of the underlying construction.

Companion material. All scripts, logs and registered readings referenced here live in the companion repository; every number below carries its provenance. Numbers that this report could not verify are not stated.

I. INTRODUCTION

Tensor networks that reproduce the geometry of hyperbolic space are a standard route from entanglement to holography [3, 4]. The MERA [5–7] is the canonical example: it is a scale-invariant ansatz whose causal cone is a cone, whose tensors are constrained to be isometries, and which is built to be optimised variationally.

Hyper-invariant tensor networks [1] propose to replace the isometry constraints on *individual* tensors with *multi-tensor* constraints on *products* of tensors, thereby allowing a network that is uniform in the bulk — a $\{7, 3\}$ tiling of the hyperbolic disk, with a three-index tensor A on every node and a matrix B on every edge. Concretely the source imposes that a product of one A and two B's acts as a $2 \to 1$ isometry w, and that a product of three A's and five B's acts as a $3 \to 2$ isometry u.

Our interest was applied, not critical. We were assembling a numerical pipeline in which a hyperbolic tensor network was to supply the bulk geometry of a holographic

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layer, and we intended to take the hyper-invariant network off the shelf. Before optimising anything we ran the elementary consistency checks that any user of a proposed ansatz should run — does the example the source prints actually satisfy the constraints the source states? It does not, and the failure is elementary enough to report on its own.

What this paper is, and is not.

- It is a *source-level* report. We read the LaTeX source of the arXiv v1 submission, not a rendering of it, and we compare tensor against tensor. Every algebraic statement below is checkable by hand from the printed matrix; the numerics merely confirm the algebra.
- It concerns *what is printed* in the specific source we read. The nature of this finding is a **literature erratum** — a concrete implementation defect in the printed tensor, *not* a critique of the method's theoretical basis. We do *not* claim that the hyper-invariant tensor network construction is wrong, that its underlying theory fails, or that any other version contains the same entry. We explicitly distinguish between (i) a specific typographical omission in one matrix entry of a v1 source snapshot, and (ii) the validity of the multi-tensor constraint framework. See Sec. X for the exact scope.
- It is *not* a claim about holography, AdS/CFT, or the physics the construction is meant to describe. Nothing here bears on whether hyper-invariant networks are useful.

Reproducibility posture. All checks are deterministic and were run against frozen source snapshots, with scripts and logs archived. We distinguish throughout between (i) exact algebraic statements, (ii) floating point readings from a stated script, and (iii) statements we could *not* verify. Nothing in category (iii) is asserted.

II. THE CONSTRUCTION AND THE CONSTRAINTS IT STATES

A. The $\{7,3\}$ network and its internal structure

The source builds the network from a $\{7,3\}$ tiling: A is a three-index tensor placed on each node, B a matrix on each edge, with A constrained to be invariant under cyclic index permutation and B to be symmetric. To exhibit a solution of the multi-tensor constraints, the source gives A and B an *internal* structure: the bond dimension χ is factorised as $\chi = \tilde{\chi}^4$, each A is built from three tensors Y_{ijkl} of bond dimension $\tilde{\chi}$, and each B is built from a unitary tensor Q_{ijkl} times a tensor R_{ijkl}.

The tensors Y are then *stated* to be *doubly unitary*, i.e. simultaneously unitary across two partitions of their four indices:

$$\sum_{kl} Y_{ijkl}\, Y^*_{mnkl} = \delta_{im}\delta_{jn}, \tag{1}$$

$$\sum_{jl} Y_{ijkl}\, Y^*_{mjnl} = \delta_{im}\delta_{kn}. \tag{2}$$

Equation (1) is unitarity in the grouping $(ij)|(kl)$ — which we write (12|34) — and Eq. (2) is unitarity in the grouping $(ik)|(jl)$, written (13|24). Following the source we call such a tensor *doubly unitary*.

B. The printed example

For a $\mathbb{Z}_2$-symmetric tensor with $\tilde{\chi} = 2$ the source prints the example, reshaped into a 4×4 matrix according to the $(ij)|(kl)$ grouping,

$$Y^{\text{orig}}_{(ij)(kl)} = \begin{pmatrix} c_1 & 0 & 0 & s_1 \\ 0 & s_1 & \boldsymbol{i}\, c_1 & 0 \\ 0 & \boldsymbol{i}\, c_1 & s_1 & 0 \\ s_1 & 0 & 0 & c_1 \end{pmatrix}, \qquad \begin{aligned} c_1 &= \cos\theta_1, \\ s_1 &= \sin\theta_1, \end{aligned} \tag{3}$$

with $\boldsymbol{i}$ the imaginary unit; the source additionally chooses Y reflection symmetric, $Y_{ijkl} = Y_{klij}$. The caption accompanying this parameterisation states that tensors built in this way "exactly satisfy the multi-tensor constraints up to (unimportant) multiplicative constants". That claim is what we checked.

III. THE PRINTED Y IS NOT DOUBLY UNITARY

A. Exact statement

The matrix (3) is real except for the two entries $\pm\boldsymbol{i}c_1$. Since it is symmetric, $Y^\dagger = \overline{Y}$, and a direct multiplication gives

$$Y^{\text{orig}}\, Y^{\text{orig}\,\dagger} = \begin{pmatrix} 1 & 0 & 0 & 2c_1 s_1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 2c_1 s_1 & 0 & 0 & 1 \end{pmatrix} = \mathbb{1}_4 + 2c_1 s_1 \left(|1\rangle\langle 4| + |4\rangle\langle 1|\right). \tag{4}$$

Every diagonal entry is $c_1^2 + s_1^2 = 1$; the only deviation from the identity is the corner pair $(1,4) = (4,1) = 2c_1 s_1$. Hence Eq. (1) fails whenever $2c_1 s_1 \neq 0$, i.e. for every θ_1 that is not a multiple of $\pi/2$:

$$\left\| Y^{\text{orig}} Y^{\text{orig}\,\dagger} - \mathbb{1} \right\|_{\max} = 2|c_1 s_1|. \tag{5}$$

The printed tensor is therefore *not* doubly unitary — indeed it is not unitary in either grouping. At the reference point $\theta_1 = 0.37$, Eq. (5) evaluates to 6.742879×10^{-1}.

Two features of Eq. (4) are worth emphasising because they shape everything that follows. First, the failure is

a *single* rank-two perturbation supported on the pair of compound indices $\{(1,1),(4,4)\}$ — it is not a diffuse numerical disagreement. Second, the offending quantity is exactly the same for both partitions of the printed tensor: the (13|24) unitarity condition (2) also fails with residual 6.742879×10^{-1}.

B. Numerics

The algebra above was confirmed numerically against the source snapshot. The (12|34) and (13|24) Gram residuals of Y^{orig} are both 6.742879×10^{-1} at the reference point, in agreement with Eq. (5). The sub-tensors required by the source's own parameterisation — R, Q and the reflection tensors of the same appendix — are *exactly* unitary, with residuals 0 and 2.220446×10^{-16} respectively. The failure is therefore specific to the single matrix (3) and is not an artefact of how we transcribed or contracted the appendix. A raw diff of our snapshot against the published `Hyper.tex` is byte-identical, so the discrepancy is in the source, not in our copy of it.

C. A single factor of i

The deviation (4) is repaired by one change: replace the four corner entries $s_1 \to \boldsymbol{i}\, s_1$,

$$Y^{\mathrm{SP}}_{(ij)(kl)} = \begin{pmatrix} c_1 & 0 & 0 & \boldsymbol{i}\, s_1 \\ 0 & s_1 & \boldsymbol{i}\, c_1 & 0 \\ 0 & \boldsymbol{i}\, c_1 & s_1 & 0 \\ \boldsymbol{i}\, s_1 & 0 & 0 & c_1 \end{pmatrix}. \tag{6}$$

Now the corner pair cancels, $\sum_k Y^{\mathrm{SP}}_{1k}\overline{Y^{\mathrm{SP}}_{4k}} = c_1\overline{\boldsymbol{i} s_1} + \boldsymbol{i} s_1 \overline{c_1} = -\boldsymbol{i} c_1 s_1 + \boldsymbol{i} c_1 s_1 = 0$, and $(Y^{\mathrm{SP}} Y^{\mathrm{SP}})^\dagger = \mathbb{1}_4$ exactly. Section V shows that Eq. (6) is not our invention: it is a tensor that has already been published independently, and it is exactly the tensor that satisfies the source's multi-tensor constraints.

IV. WHAT A CONTRACTION CAN SEE: (12|34) IS THE ONLY PARTITION IN PLAY

The observation that a single factor of i both repairs the tensor and recovers the published one would be a curiosity if *both* unitarity conditions mattered. They do not. We now show that the contractions defining w and u can only ever test one of the two partitions, so that the source's doubly-unitary requirement is strictly stronger than its own constraints require.

A. The Gram identity for w

Write $A[a,m,n]$, expanding the leg a into the rows of Y_1 and Y_2, the leg m into the columns of Y_1 and the rows of Y_3, and the leg n into the columns of Y_2 and Y_3; build $w = A \cdot B \cdot B$ with the two B's contracted against each other. Contracting the pair of B's first collapses them to a Kronecker delta, leaving

$$(WW^\dagger)[a,a'] = \sum_{mn} A[a,m,n]\, A^*[a',m,n], \tag{7}$$

and evaluating Eq. (7) with the leg-map above gives the identity

$$\boxed{WW^\dagger \;=\; \mathrm{Tr}\big(Y^\dagger Y\big)\,\Big(P_{12}(Y_1)\otimes P_{12}(Y_2)\Big)} \tag{8}$$

where $P_{12}(Y)$ is the (12|34) Gram matrix of a single Y,

$$P_{12}(Y)_{(ij),(mn)} \;=\; \sum_{kl} Y_{ijkl}\, Y^*_{mnkl}. \tag{9}$$

The derivation is one line of index bookkeeping and is reproduced in Appendix A: in Eq. (7) the tensors Y_1 and Y_2 each contribute one free (12|34) Gram, while the third copy Y_3 is fully contracted and contributes only the scalar $\mathrm{Tr}(Y^\dagger Y)$.

Equation (8) is the whole story. Because $P_{12} \otimes P_{12}$ is a direct product and the prefactor is a positive scalar,

$$w \text{ co-isometric} \iff P_{12}(Y) \propto \mathbb{1}_4 \iff Y \text{ unitary on } (12|34), \tag{10}$$

with normalisation $c = \mathrm{Tr}(Y^\dagger Y)\big(\mathrm{Tr}P_{12}/4\big)^2$. The other two groupings (13|24) and (14|23) have *no* corresponding free leg in the leg-map of A; they cannot appear. This is the mechanism behind the title of this section. Note also that Eq. (8) is manifestly independent of the surrounding B's beyond the overall constant: $P_{12}(Y)$ is a factor of $WW^\dagger$ in its own right.

B. The same condition governs u

The $3 \to 2$ constraint $u = A^3 B^5$ defines a U whose co-isometry condition factorises similarly. Its four 4×4 factors are $\propto \mathbb{1}$ with $F_a = P_{12}(Y)$; the quantities F_a are *direct-product factors of* $UU^\dagger$, not merely proportional to it, and are independent of B. Hence

$$u \text{ co-isometric} \Longrightarrow F_a = P_{12}(Y) \propto \mathbb{1}_4 \Longrightarrow Y \text{ unitary on } (12|34), \tag{11}$$

the *same* law as Eq. (10). The two multi-tensor constraints are therefore not two conditions but one.

C. Consequence: (13|24) is redundant

Combining (10) and (11): the conjunction of the source's multi-tensor constraints is equivalent to unitarity of Y on the *single* partition (12|34). The second half of the doubly-unitary requirement, Eq. (2), is never exercised by the constraints the source writes down. A tensor

that is unitary on (12|34) and *not* on (13|24) satisfies w and u exactly.

This is not a technicality, because the printed tensor (3) fails precisely the partition that matters and the published tensor (6) satisfies precisely it. The repair in Sec. III is therefore not cosmetic: it restores the necessary condition.

D. Numerical confirmation

We confirmed Eqs. (8)–(11) numerically.

- *Identity* (8): residuals at the relative 10^{-16} level, i.e. at the floating-point noise floor of the contraction.
- *Sufficiency*: random 4×4 unitaries (with (13|24) and (14|23) both broken) yield exact w and u and $c = 4.0000$, in 8/8 draws.
- *Necessity*: tensors unitary *only* on (13|24) fail, 3/3, with w in 2.74–2.83; tensors unitary *only* on (14|23) fail, 3/3, with w in 2.71–3.71.
- *Scaling*: for $Y = \lambda U$ with U unitary, the measured constants are $c_w = 4\lambda^6$ and $c_u = 4\lambda^{18}$ to the printed digits, with a log–log slope of 18.0000. The exponent 18 counts nine copies of Y, each contributing one quadratic factor — structurally the same bookkeeping as the "two (12|34) Gram factors plus one Frobenius norm per A" of Eq. (8).

Finally, the B-independence of Eq. (8) was checked directly by recomputing $F_a = P_{12}(Y)$ against three different B constructions, including two *non-unitary* ones taken from the alternative factorisations discussed in Sec. V: F_a is 6.743×10^{-1} for Y^{orig} under all three B, bit-for-bit, and $\leq 3.3 \times 10^{-16}$ for Y^{SP} under all three.

V. AN INDEPENDENT PUBLISHED WITNESS

Equation (6) is not a repair we invented. Steinberg and Prior [2] publish a doubly-unitary-style tensor for the same construction, and their printed matrix differs from Eq. (3) in exactly the four corner entries: where the source prints the *real* s_1, they print the *imaginary* $\boldsymbol{i}s_1$. We verified their tensor numerically: its (12|34) Gram residual is 0, its (13|24) residual is 6.742879×10^{-1}, and — the point of this section — it satisfies the source's multi-tensor constraints.

The contrast is sharp and is the clearest single statement of the finding:
(The w and u residuals are against the source's own multi-tensor constraints; the corrected tensor additionally gives $c = 4$ exactly.) Two conclusions follow.

(a) The defect is a single missing factor of i. The only difference between a tensor that fails the constraints by $O(1)$ and one that satisfies them to machine precision is $s_1 \to \boldsymbol{i}s_1$ in four entries. This is consistent with a typographical omission, and we state it as such: we make no claim about intent. **Crucially, this observation is an erratum concerning a specific printed tensor, not a refutation of the hyper-invariant method. The theoretical framework remains untouched; only the concrete example in the v1 source requires correction.**

(b) A reusable admissibility criterion. Combining Sec. IV with the table above,

$$Y \text{ unitary on } (12|34) \iff \text{the multi-tensor constraints hold} \iff \text{t} \tag{12}$$

Because (12|34) unitarity is the *necessary* condition, a candidate solution of the multi-tensor constraints can be screened by a single 4×4 computation, without contracting any network. Conversely, the source's demand that Y be doubly unitary rejects tensors — such as the published (6) — that in fact satisfy every constraint the source writes down.

A text–matrix tension in the published witness. We note in passing that the text of Steinberg and Prior [2] describes its Y and R as restricted to being "doubly unitary", whereas the matrix it prints is unitary on one of the two partitions only ($\mathrm{dev}(13|24) = 6.742879 \times 10^{-1}$). By the result of Sec. IV this is harmless — the second partition is never tested — but it does mean that the commonly quoted statement "Y is doubly unitary" is, for both of the works we examined, not what the printed matrices show.

VI. THE MULTI-TENSOR CONSTRAINTS THEMSELVES

Equation (12) reduces the source's constraints to a property of Y alone. It is worth recording the direct check of the constraints as written, since it localises the failure.

Two readings are available and we register both. Read one way, B being exactly unitary shows that the failure is not a global normalisation problem but is carried entirely by Y. Read another way — and this matters — B's unitarity is *not* a precondition: the published alternative factorisations of Steinberg and Prior [2] replace R by tensors that are antisymmetric and explicitly non-unitary, and we verified that the Y-side failure is unchanged in that setting. $F_a = P_{12}(Y)$ is 6.743×10^{-1} for the printed Y under a unitary B, under a non-unitary B, and under a third construction, bit-for-bit. This is the content of the B-independence asserted in Sec. IV.

tensor	dev(12|34)	dev(13|24)	w resid.	u resid.
printed in `Hyper.tex` ((3))	6.742879×10^{-1}	6.742879×10^{-1}	2.697152	9.497650
published [2] ((6))	0	6.742879×10^{-1}	1.776357×10^{-15}	3.552714×10^{-15}

constraint	printed (3)	corrected (6)
B (does not contain Y)	exactly unitary	exactly unitary
$w = A \cdot B \cdot B$, $2 \to 1$ isometry	2.697152 (fail)	1.776357×10^{-15}
$u = A^3B^5$, $3 \to 2$ isometry	9.497650 (fail)	3.552714×10^{-15}

quantity	value
vertices / edges / faces	$V = 2240$, $E = 2856$, $F = 617$
layer identity	$\max \lvert\text{err}\rvert = 4.918 \times 10^{-16}$
$g_k(Z_k) = Z_{k+1}$, $g_k(Z_{k+1}) = Z_k$	
Euler characteristic	$V - E + F = 1$
connectivity	single component
plaquettes	every face has 7 edges and 7 vertices
edge gluing	1393 boundary / 1463 interior, no edge shared by ≥ 3 faces
vertex valence	all 847 interior vertices of valence 3

VII. A SECOND BUILDING BLOCK: TENSOR NETWORK RENORMALIZATION

The same genre of check, applied to a different advertised building block, gives a second result. We examined the appendix of arXiv:1412.0732 (`TNR.tex`), Evenbly and Vidal [8], which factorises a coarse-graining tensor. Of the two factorisation identities printed there,

- the first (`Eq.(TRG2)`) equates a tensor built from a real tensor S with a sum of Gram matrices of real vectors, hence with a *positive-semidefinite* object;
- the second (`Eq.(TRG4)`) constrains the same tensor differently.

Enumerating leg orderings, *all* 24/24 admissible orderings of the printed tensor give a matrix that is *not* positive semidefinite, the minimum eigenvalue being -2.0. The first identity is therefore impossible for any real S, independently of index ordering or pairing convention. The only reading that can be satisfied is the second; but of 384 enumerated readings of that equation, the 12 that work do not contain the printed S. The two identities are mutually exclusive as printed.

We report this because it constrains what a user may take off the shelf: the factorisation cannot be imported as written, and a working implementation must supply its own.

VIII. A CONSTRUCTION THAT DOES HOLD: THE $\{7, 3\}$ LAYER GEOMETRY

It is worth recording that the structural half of the hyper-invariant construction fared well, because it bounds the failure. The network is organised into concentric layers about a bulk point, and each layer is a mapping between a lattice $\mathcal{L}_z$ and a coarser $\mathcal{L}_{z+1}$. We built the $\{7, 3\}$ tiling combinatorially and assembled the layer from its 2-site and 3-site unit cells. The finite patch is combinatorially self-consistent:

Two remarks. First, the layer assembly requires only the (12|34) property of Y — consistent with Sec. IV and closing off any residual worry that the alternative grouping might be needed elsewhere. Second, the scale factor of the tiling works out as expected: the layer populations follow $N(\mathcal{L}_z) = F(2z + 5)$ (Fibonacci) and their ratios approach $\varphi^2 = 1 + \varphi = 2.618\ldots$ monotonically, matching the source's stated scale factor for the $\{7, 3\}$ tiling. We record two gaps rather than smoothing them over: our layer is an *abstract* cell string, not a layer *geometrically extracted* from the tiling (the extraction rule is not printed in the source); and we did not perform the full-layer contraction, which is exponentially expensive.

IX. OUR OWN BUILDING BLOCKS: MERA AND CMERA

For symmetry we report the corresponding checks on the two tensor-network blocks that we built ourselves, since a reader is entitled to ask whether the same scrutiny was applied inward.

A. MERA

Our MERA is exactly isometric *by construction* (it is QR-parameterised), with residuals at the 10^{-16} level; this is an identity, not a fit, and we do not present it as evidence of anything beyond correctness of implementation. It is also a pure tree: the triangle count of the MERA graph is 0, so the triangulated Forman curvature reduces to the tree form $F = 4 - \deg(u) - \deg(v)$, which we therefore always read alongside the triangle count rather than alone.

The informative result is a *negative control*. Injecting a non-unitary perturbation of strength ε into the MERA tensors, on an isolated path so that no registered metric is disturbed, we find two things that must be taken together.

- *A large gap.* The isometry residual crosses its 10^{-12} threshold at $\varepsilon = 10^{-6}$, whereas the downstream central-charge reading `eps_c` [9] degrades only at $\varepsilon = 3 \times 10^{-2}$. The two scales are separated by

$\sim 3 \times 10^4$ — the isometry check is about four orders of magnitude more sensitive than the downstream observable.

- *Non-monotonicity.* The degradation is *not* monotone in ε: at $\varepsilon = 10^{-3}$ the downstream reading is *below* its unperturbed baseline, at $0.787\times$ it. This forbids the tempting reading "small perturbations are harmless"; it says instead that the downstream quantity has a noise floor.

We also found that a second candidate downstream quantity, R^2_{linear}, is 0.800000 identically across all eight values of ε tested — it counts tensors and is structurally incapable of being a sensitive probe. Neither result promotes the layer's maturity; both are registered as negative controls with teeth.

B. Gaussian cMERA

Our Gaussian cMERA [10] passes its self-check 9/9, and we regard that check as valuable for a methodological reason: the criteria were fixed *before* the convention was chosen, so that the convention had to satisfy the criteria rather than the reverse — exchanging the convention fails the check. That is what a self-check should do.

It does not, however, establish that the object carries physical content, and here the checks were negative. A probe of the scale-invariance observable returns `EXIT=3`: in the Gaussian convention we adopt there are no discrete layers and hence no per-layer generator, so the original question has no observable to compute — this is a "criterion not applicable", not a regression. More decisively, a discriminating-power probe at $L = 8$, dim $= 256$ shows that the norm $\|L|\Omega\rangle\| = 2.915476$ is *bit-identical* across the ultraviolet vacuum, the cMERA flow end-state and the transverse-field Ising ground state [11], with a relative spread of 2.285×10^{-15}. On that family no observable of the form $L|\Omega\rangle$ can separate the cMERA end-state from the bare ultraviolet vacuum. The block is therefore not entered into the layer's scored path, and the layer's maturity is unchanged.

We report this as a boundary on our own work, not as a result about cMERA in general: it is scoped to the convention we adopt and the family we probe.

X. SCOPE AND BOUNDARIES

We state exactly what the claims do and do not cover.

(1) Version scope. All statements about the hyper-invariant construction are about the *arXiv v1 source* arXiv:1704.04229v1, whose `Hyper.tex` we read and against which our snapshot is byte-identical. That arXiv entry has a single version (no v2/v3), so a version comparison cannot answer whether a later published version changed the entry. The published journal version was not available to us and we therefore make *no claim* about it. Readers with access should check whether the printed Y in that version carries $\boldsymbol{is}_1$ in the corners.

(2) Claim scope and nature of the finding. We claim that the printed tensor in the arXiv v1 source does not satisfy the printed constraint. **The proper characterization of this finding is a *literature erratum* — a defect in a specific implementation (the printed 4×4 matrix), not a failure of the method's theoretical foundation.** We do *not* claim that the hyper-invariant tensor network construction is invalid, that its underlying theory is incorrect, or that the authors' other results fail. The single-factor-of-i repair, and the existence of an independent published tensor carrying exactly that factor, are consistent with a typographical omission; we record that as an observation, not a finding about motivation. A reader should understand this paper as *correcting a concrete printing error, not as challenging the merit of the hyper-invariant approach.*

(3) What would change the conclusion. A published version whose Y has $\boldsymbol{is}_1$ in the corners would move the defect from "source" to "version", and the (12|34) theorem would be unaffected either way — it is a statement about the contractions, not about any particular tensor.

(4) Unverified items. We did not verify the published journal versions of either work; we did not attempt the full-layer contraction of the hyper-invariant network; and the "extraction rule" for the geometric layer is not printed in the source and was replaced by an exhaustive substitution family in our implementation. These gaps are registered, not smoothed over.

(5) On the two readings of B. We deliberately registered two readings of the multi-tensor result rather than choosing one, because the choice changes what the result means for a user: if B's unitarity is a precondition, the failure is localised to Y; if it is not (as the published alternative factorisations show), the failure is a property of Y alone that survives changes of the surrounding tensors. Both readings are in Sec. VI.

(6) Relation to original authors' work. We have taken care throughout to frame this report as a source-level *erratum*. The missing factor of i is a defect in *what was printed*, not in *what was intended*. We do not attribute error to the authors' reasoning, methodology, or theoretical contributions. Our expectation is that a corrected tensor (e.g., as in Steinberg and Prior [2]) resolves the constraint failure entirely, leaving the hyper-invariant framework intact. This paper is offered in the spirit of scholarly correction, not criticism.

XI. REPRODUCIBILITY

Every number in this paper was produced by a deterministic script run against a frozen source snapshot,

with the script and its log archived. Algebraic statements, Eqs. (4)–(11) and (12), are checkable by hand and are confirmed numerically; the numerical confirmations are stated as readings with their magnitudes, so that a reader can judge the significance rather than the digit count. Source snapshots (`Hyper.tex`, `TNR.tex`) are archived with the scripts. The reference point for the Y checks is $\theta_1 = 0.37$; where a reading depends on a convention (in particular the Gaussian cMERA convention) the convention is named beside the reading.

A note on numbers we did not use. Our archive contains readings whose recorded run times could not be reconciled with each other. Following our own house rule we register such discrepancies without explaining them, and none of them enters any conclusion here.

Appendix A: Proof of the Gram identity (8)

Let $A[a, m, n]$ be the A tensor, with leg a expanded into the rows of Y_1 and Y_2, leg m into the columns of Y_1 and the rows of Y_3, and leg n into the columns of Y_2 and Y_3. Form $w = A{\cdot}B{\cdot}B$ by contracting the pair of B's against each other; with B unitary the pair collapses to δ. Then

$$(WW^\dagger)[a, a'] = \sum_{mn} A[a, m, n]\, A^*[a', m, n]. \tag{A1}$$

Expanding A into its three Y factors and performing the sums over the finer indices, the copies Y_1, Y_2 each close into a (12|34) Gram, Eq. (9), because their free indices are precisely the ones left open in a and n; the third copy Y_3 has no free leg and closes into the scalar $\sum_{ijkl} |Y_{ijkl}|^2 = \mathrm{Tr}(Y^\dagger Y)$. Hence

$$WW^\dagger = \mathrm{Tr}(Y^\dagger Y)\,\big(P_{12}(Y_1) \otimes P_{12}(Y_2)\big), \tag{A2}$$

which is Eq. (8). Since the right-hand side is a direct product of two 4×4 factors with a scalar prefactor, it is proportional to the identity iff $P_{12}(Y) \propto \mathbb{1}_4$, giving Eq. (10). The groupings (13|24) and (14|23) produce no such factor because the corresponding index pairs are not simultaneously free on any single leg of A. The same expansion applied to $u = A^3B^5$ yields $F_a = P_{12}(Y)$ as a direct-product factor of $UU^\dagger$, and the factor is independent of B because B's contractions act on the other legs; this is Eq. (11). □

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